

GASSIM, Saudi ArabiaWMO: **404050**Lat: **26.300N**Lon: **43.767E**Elev: **648**StdP: **93.78**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.0	5.9	N/A	N/A	N/A	N/A	N/A	N/A	9.5	15.8	8.4	16.7	1.3	30	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.1	45.2	N/A	44.2	N/A	43.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	8.0	7.0	0.5	46.6	1.9	0.9	-0.9	47.2	-2.0	47.8	-3.1	48.3	-4.5	48.9		
			DB													
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	25.9	13.9	16.4	20.5	25.8	31.4	34.3	35.4	35.8	33.0	27.8	20.3	15.4	(6)
		DBStd	8.45	3.44	3.85	3.77	3.34	2.83	1.79	1.94	1.82	2.17	2.53	3.62	3.38	
		HDD10.0	11	7	2	0	0	0	0	0	0	0	0	0	3	
		HDD18.3	364	142	76	21	1	0	0	0	0	0	0	22	103	
		CDD10.0	5808	128	181	326	474	664	730	787	800	691	551	308	169	
		CDD18.3	3120	5	22	89	224	406	480	528	542	441	292	80	11	
		CDH23.3	49774	66	255	991	2716	6292	8378	9492	9671	7224	3818	763	107	
(13)		CDH26.7	33967	8	66	406	1451	4154	6052	7052	7221	5024	2239	281	13	(13)
(14)	Wind	WSAvg	3.1	2.9	3.3	3.6	3.8	3.6	3.3	3.1	2.9	2.6	2.6	3.0	2.8	(14)
(15)	Precipitation	PrecAvg	106	13	12	18	18	5	0	0	0	4	23	13		(15)
(16)		PrecMax	247	93	57	80	47	41	9	1	0	4	34	113	105	(16)
(17)		PrecMin	24	0	3	0	0	0	0	0	0	1	2	0		(17)
(18)		PrecStd	56	21	14	22	16	9	2	0	1	7	28	24		(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	27.2	31.2	36.0	39.3	43.5	45.2	46.8	46.2	44.8	39.8	33.8	28.2		(19)
(20)		0.4% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(20)
(21)		2% DB	25.1	28.8	33.2	37.8	42.1	44.1	45.2	45.2	43.2	38.2	32.0	26.1		(21)
(22)		2% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(22)
(23)		5% DB	23.0	26.6	31.0	35.8	40.8	43.1	44.2	44.8	42.1	37.1	30.0	24.1		(23)
(24)		5% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(24)
(25)		10% DB	20.9	24.2	28.8	33.9	39.1	42.1	43.2	43.8	41.0	35.9	28.0	22.1		(25)
(26)		10% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(27)
(28)		0.4% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(28)
(29)		2% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(29)
(30)		2% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(30)
(31)		5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(31)
(32)		5% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(32)
(33)		10% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(33)
(34)		10% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(34)
(35)	Mean Daily Temperature Range	MDBR	12.1	13.3	13.8	14.1	14.6	15.8	16.0	16.1	16.2	15.5	12.6	12.1		(35)
(36)		5% DB MCDBR	15.0	15.9	16.8	16.0	15.7	16.4	16.2	16.5	16.8	16.7	15.5	14.6		(36)
(37)		5% DB MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(37)
(38)		5% WB MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(38)
(39)		5% WB MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.417	0.460	0.535	0.596	0.626	0.566	0.537	0.566	0.530	0.489	0.488	0.434		(40)
(41)		taud	1.999	1.878	1.705	1.619	1.566	1.667	1.754	1.693	1.747	1.826	1.835	1.954		(41)
(42)		Ebn at Noon	810	805	768	734	712	751	772	748	761	765	727	769		(42)
(43)		Edh at Noon	154	188	236	264	278	250	228	242	223	195	181	155		(43)
(44)	All-Sky Solar Radiation	RadAvg	4.15	5.04	6.00	6.54	7.29	8.06	7.82	7.31	6.57	5.56	4.21	3.84		(44)
(45)		RadStd	0.28	0.26	0.26	0.31	0.33	0.23	0.20	0.18	0.19	0.24	0.35	0.24		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.81	N/A	N/A	+0.77	N/A	N/A	N/A	-57	+326	+272
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page